

WHAT YOU WILL STUDY IN TODAY VIDEO ?

- ▶ CNN Backpropagation
- ▶ Easy Diagram Based Working

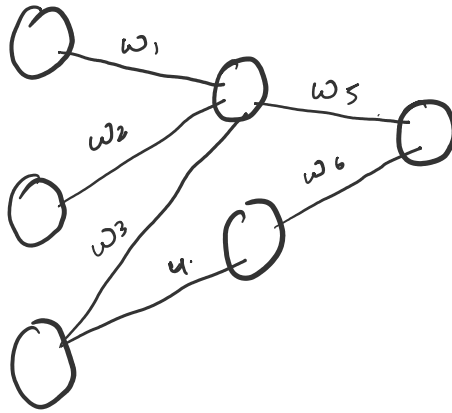
Prerequisites

- ① ANN/MLP Backprop.
- ② Convolution layer. //

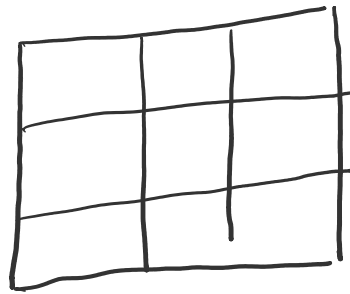
Important

Aim of Backprop → Weight updation

ANN



CNN



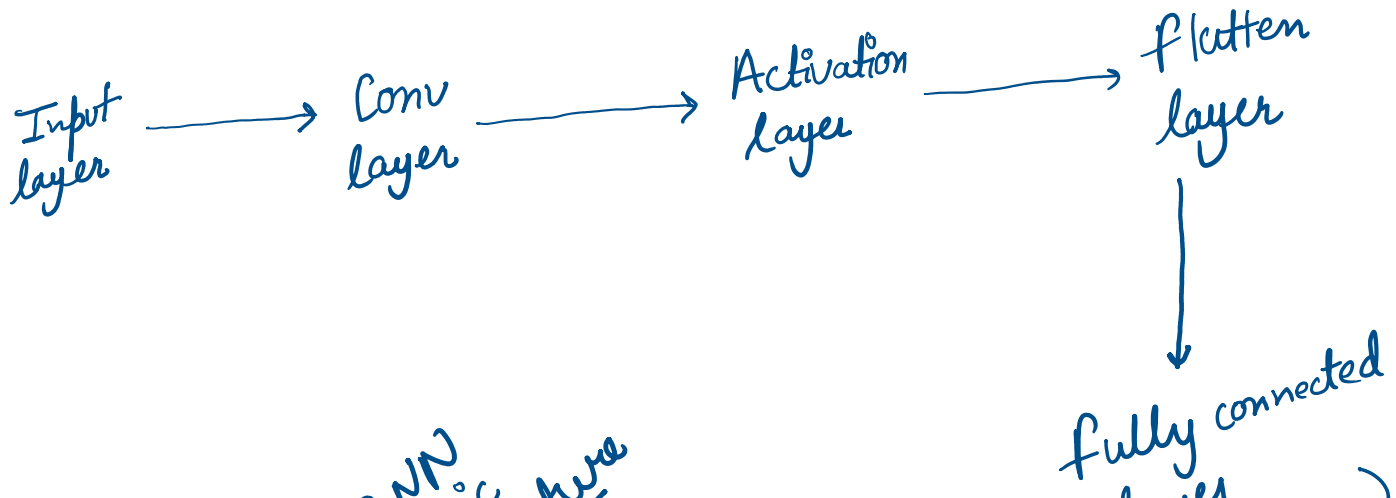
3x3
image

w_1	w_2
w_3	w_4

2x2
filter / kernel

update

CNN Model



CNN
Basic
Architecture

fully con.
layer
(ANN layer)

Example.

① Input layer (5x5 image)
25 pixels

a_1	a_2	a_3	a_4	a_5
a_6	a_7	a_8	a_9	a_{10}
a_{11}	a_{12}	a_{13}	a_{14}	a_{15}
a_{16}	a_{17}	a_{18}	a_{19}	a_{20}
a_{21}	a_{22}	a_{23}	a_{24}	a_{25}

Kernel. (3x3)
(9 weights)
update.

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

② forward propagation

Convolution operation

Stride = 2.

Weighted Sum (Z)

a_1	a_2	a_3
a_6	a_7	a_8
a_{11}	a_{12}	a_{13}

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$Z_1 = a_1 w_1 + a_2 w_2 + a_3 w_3 + \dots + a_{13} w_9$$

Z_2
 Z_3
 Z_4

Do same

feature map.

Z_1	Z_2
Z_3	Z_4

Activation layer (ReLU)

→ $\max(0, x)$

$\max(0, Z_1)$

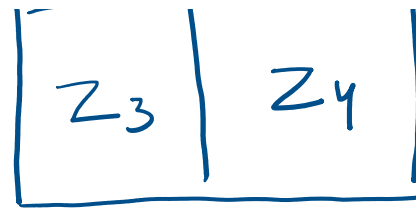
feature map

→

Z_1	Z_2
Z_3	Z_4

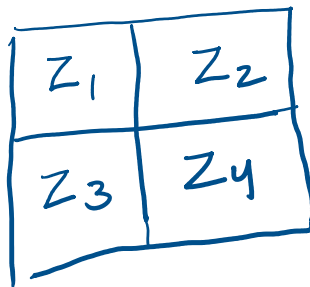
→

few
map
(same as
all were +ve)

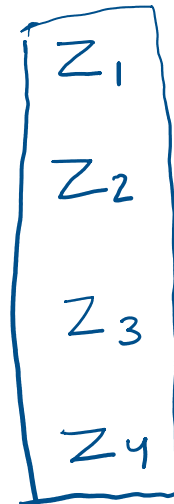


(2-D)

flatten
layer

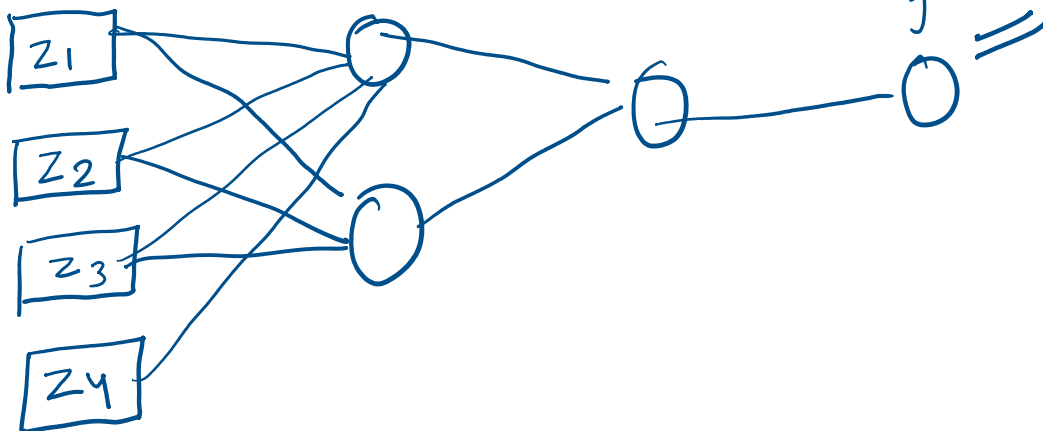


(2x2)
(2-D)



4x1= (1-D)

Dense
layers



Predicted
value

③ loss calculation

Compare True value (y) & Predicted value $(\hat{y})$

You will get loss (L)

Get those weights where loss is minimum.

loss minimize

→ Optimizers
(Gradient Descent)

(Loss) ↓ ↓ ↓ ↓

Backpropagation

Weight updation formula

$$w_{\text{new}} = w_{\text{old}} - \underset{\substack{\downarrow \\ \text{learning} \\ \text{rate}}}{\alpha} \cdot \underset{\substack{\downarrow \\ \text{loss gradient} \\ \text{wrt} \\ \text{weight}}}{\frac{dL}{dw}}$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

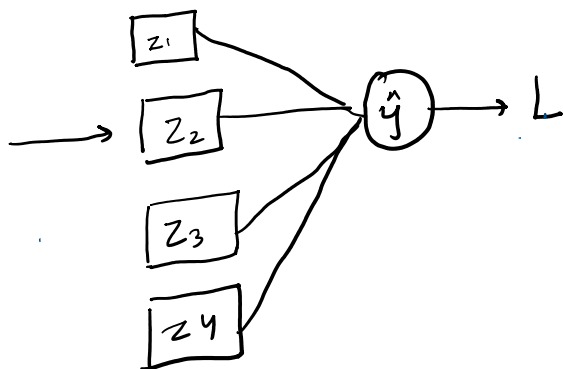
weight matrix

$\frac{dL}{dw_1}$	$\frac{dL}{dw_2}$	$\frac{dL}{dw_3}$
$\frac{dL}{dw_4}$	$\frac{dL}{dw_5}$	$\frac{dL}{dw_6}$
$\frac{dL}{dw_7}$	$\frac{dL}{dw_8}$	$\frac{dL}{dw_9}$

Derivatives & chain rule operation ← loss gradient matrix

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

z_1	z_2
z_3	z_4



lets take w_1

$$\frac{dL}{dw_1} = \frac{dz_1}{dw_1} \frac{dL}{dz_1} + \frac{dz_2}{dw_1} \frac{dL}{dz_2} + \frac{dz_3}{dw_1} \frac{dL}{dz_3} + \frac{dz_4}{dw_1} \frac{dL}{dz_4}$$

Why used all z for this $\frac{dL}{dw_1}$?

$$\begin{aligned} z_1 &= w_1 a_1 + w_2 a_2 \\ z_2 &= w_1 a_3 + w_2 a_4 \\ z_3 &= w_1 a_{11} + w_2 a_{12} \\ z_4 &= w_1 a_{13} + w_2 a_{14} \end{aligned}$$

Change in w_1 is affect z_1, z_2, z_3, z_4

Equation

$$\frac{dL}{dw_1} = \frac{dz_1}{dz_1} \underbrace{\frac{dz_1}{dw_1}}_{a_1} + \frac{dz_2}{dz_2} \underbrace{\frac{dz_2}{dw_1}}_{a_3} + \frac{dz_3}{dz_3} \underbrace{\frac{dz_3}{dw_1}}_{a_{11}} + \frac{dz_4}{dz_4} \underbrace{\frac{dz_4}{dw_1}}_{a_{13}}$$

New
equation

$$\frac{dL}{dw_1} = a_1 \frac{dL}{dz_1} + a_3 \frac{dL}{dz_2} + a_{11} \frac{dL}{dz_3} + a_{13} \frac{dL}{dz_4}$$

//

You have to
get same eqn
for all
q weight



q eqns.

9 eqns

$$\frac{dL}{d\omega_1} = \overset{\textcircled{1}}{a_1 \frac{dL}{dz_1}} + \overset{\textcircled{2}}{a_3 \frac{dL}{dz_2}} + \overset{\textcircled{3}}{a_{11} \frac{dL}{dz_3}} + \overset{\textcircled{4}}{a_{13} \frac{dL}{dz_4}}$$

$$\frac{dL}{d\omega_2} = a_2'' + a_4'' + a_{12}'' + a_{14}''$$

$$\frac{dL}{d\omega_3} = a_3'' + a_5'' + a_{13}'' + a_{15}''$$

$$\frac{dL}{d\omega_4} = a_6'' + a_8'' + a_{16}'' + a_{18}''$$

$$\frac{dL}{d\omega_5} = a_7'' + a_9'' + a_{17}'' + a_{19}''$$

$$\frac{dL}{d\omega_6} = a_8'' + a_{10}'' + a_{18}'' + a_{20}''$$

$$\frac{dL}{d\omega_7} = a_{11}'' + a_{13}'' + a_{21}'' + a_{23}''$$

$$\frac{dL}{d\omega_8} = a_{12}'' + a_{14}'' + a_{22}'' + a_{24}''$$

$$\frac{dL}{d\omega_9} = a_{13}'' + a_{15}'' + a_{23}'' + a_{25}''$$

$$\overset{\textcircled{1}}{\begin{bmatrix} a_1 & a_2 & a_3 \\ a_6 & a_7 & a_8 \\ a_{11} & a_{12} & a_{13} \end{bmatrix}} \times \frac{dL}{dz_1} + \overset{\textcircled{2}}{\begin{bmatrix} a_3 & a_4 & a_5 \\ a_8 & a_9 & a_{10} \\ a_{13} & a_{14} & a_{15} \end{bmatrix}} \times \frac{dL}{dz_2} + \overset{\textcircled{3}}{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{16} & a_{17} & a_{18} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}} \times \frac{dL}{dz_3} + \overset{\textcircled{4}}{\begin{bmatrix} a_{13} & a_{14} & a_{15} \\ a_{18} & a_{19} & a_{20} \\ a_{23} & a_{24} & a_{25} \end{bmatrix}} \times \frac{dL}{dz_4}$$

After all calculation

$$\frac{dL}{dw}$$

$\frac{dL}{dw_1}$	$\frac{dL}{dw_2}$	$\frac{dL}{dw_3}$
$\frac{dL}{dw_4}$	$\frac{dL}{dw_5}$	$\frac{dL}{dw_6}$
$\frac{dL}{dw_7}$	$\frac{dL}{dw_8}$	$\frac{dL}{dw_9}$

has grad matrix

Weight update

$$w_{new} = w_{old} - \alpha \frac{dL}{dw}$$

$$w_{new} = \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix} - 0.1 \begin{bmatrix} \frac{dL}{dw_1} & \frac{dL}{dw_2} & - \\ - & - & - \\ - & - & \frac{dL}{dw_9} \end{bmatrix}$$

Result

$w_{1_{new}}$	$w_{2_{new}}$	-
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New updated weight

$w_{1_{new}}$	$w_{2_{new}}$	-
-	-	-
-	-	$w_{9_{new}}$

→ weight matrix

